
Text-Guided AVIGATE: Query-Aware Audio-Visual Fusion for Text-to-Video Retrieval

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Abstract

Text-to-video retrieval requires matching a natural-language query to both visual and audible video cues. AVIGATE is used as the audio-visual baseline: although it selectively fuses audio and visual features, its gates are computed from the video streams alone and therefore cannot determine whether a sound is relevant to the current query. This report introduces **Text-Guided AVIGATE**, a query-aware extension with two complementary components. First, a **Text-Conditioned Gating Function** pools visual, audio, and text embeddings and predicts g_{mha} , g_{ffn} , and g_{query} from their joint context. Second, a **Text-Injected MHA Query** projects the text embedding into the visual query and uses g_{query} to control the injection strength. This design separates the decision of how much audio to fuse from the decision of which audio evidence to retrieve for a given text query. On MSR-VTT, the proposed method improves Recall@1 by **+15.8** over AVIGATE and **+9.0** over GAIS under the same backbone/modalities, with consistent gains on CLIP ViT-B/16 and VATEX.

1 Introduction

Text-to-video retrieval ranks a gallery of videos by relevance to a free-form text query. The dominant paradigm builds on large-scale image-text pretraining such as CLIP [9] and adapts it to the temporal domain, as in CLIP4Clip [7]. This paradigm is effective but often *vision-centric*: a video is treated mainly as a sequence of frames, while dialogue, sound effects, and music are discarded or used only weakly despite carrying semantics with no visual counterpart.

Audio-visual retrieval methods attempt to close this gap by adding an audio stream to the visual representation. AVIGATE [4] is used in this report as a strong audio-visual baseline because its Gated Fusion Transformer selectively injects audio into visual features. However, its fusion is **text-agnostic**: gates are computed from the video streams alone and never see query text T . This matters because audio relevance is not an intrinsic property of a clip. A bark is decisive for “a dog barking” but may be irrelevant for “a dog running on grass”; applause is useful for “people clapping” but distracting for a query about clothing or camera motion. Query-agnostic fusion can therefore waste the audio modality or inject signal that hurts the final ranking.

This report makes the AVIGATE fusion block query-aware with two components: a *Text-Conditioned Gating Function*, which predicts fusion gates from pooled visual, audio, and text embeddings, and a *Text-Injected MHA Query*, which projects text into the visual cross-attention query with strength controlled by g_{query} . The first component decides how strongly audio should be kept in the fusion block, while the second changes the retrieval content of the cross-attention query itself. Experiments on MSR-VTT [12] and VATEX [11] show that these two forms of text conditioning are complementary.

2 Method

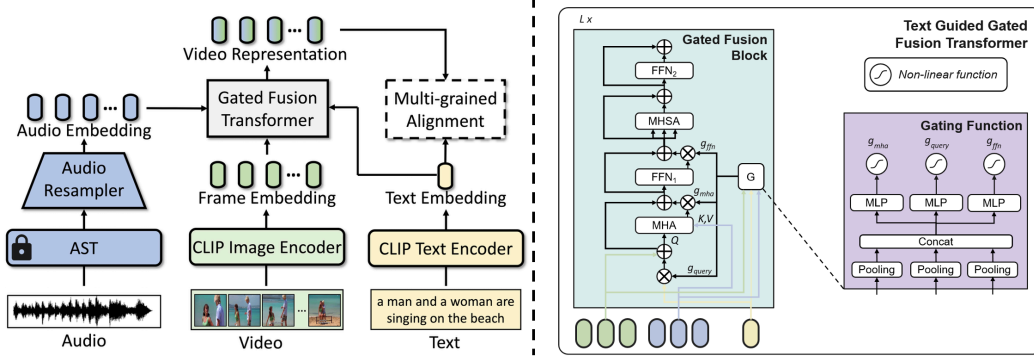


Figure 1: Overall architecture. The gating MLP predicts g_{mha} , g_{ffn} , and g_{query} from pooled visual, audio, and text features; g_{query} scales text injection into the visual query.

Architecture overview. Figure 1 summarizes the proposed pipeline. Video frames are encoded with a CLIP image encoder to obtain frame-level visual embeddings $V = \{v_i\}_{i=1}^{T_v}$, and the text query is encoded with the CLIP text encoder to obtain t . The audio waveform is represented by an audio encoder and resampled into audio tokens $A = \{a_j\}_{j=1}^{T_a}$ that can be aligned with the visual stream. The Gated Fusion Transformer then updates the video representation by attending from visual queries to audio keys and values. The main change is that the fusion block is no longer driven only by V and A : the text embedding is used both to predict the fusion gates and to modify the attention query.

Background: AVIGATE. AVIGATE fuses V and A with visual queries and audio keys/values. In a standard cross-attention view, visual tokens produce Q , audio tokens produce K and V_A , and the audio-attended feature is added back to the visual stream through gated residual connections. The important limitation is that the original gates depend only on video content, i.e. $g = f(V, A)$, independent of t . This design can learn whether an audio segment is compatible with nearby visual frames, but it cannot ask whether the audio is useful for the particular sentence being searched.

Contribution 1: Text-Conditioned Gating Function. As shown in Figure 1, pooled visual, audio, and text features are concatenated and passed to the gating MLP:

$$[g_{mha}, g_{ffn}, g_{query}] = \sigma(\text{MLP}([\text{Pool}(V) \parallel \text{Pool}(A) \parallel \text{Pool}(t)])), \quad (1)$$

where $\parallel$ is concatenation and σ denotes a sigmoid gate. g_{mha} modulates the cross-modal attention residual, and g_{ffn} modulates the feed-forward residual in the fusion block:

$$Z = V + g_{mha} \odot \text{MHA}(Q, K_A, V_A), \quad H = Z + g_{ffn} \odot \text{FFN}(Z). \quad (2)$$

The gates are broadcast over token positions when needed. Compared with AVIGATE’s video-only gate, this function gives the model direct access to query semantics before it decides how much fused audio should pass through each fusion layer. The third output, g_{query} , is not an attention or FFN gate; it controls the amount of text injected into the cross-attention query in Contribution 2.

Contribution 2: Text-Injected MHA Query. The original attention query is purely visual ($Q \leftarrow V$). Text-Guided AVIGATE instead projects text to the visual space and injects it before query projection:

$$\tilde{V} = V + g_{query} \odot \phi(t), \quad Q = W_Q \tilde{V}, \quad (3)$$

where $\phi(\cdot)$ projects text to the visual dimension. Thus g_{query} dynamically controls how much text affects the cross-attention query. A large g_{query} makes audio attention more semantic when the query names an audible event, while a small value keeps the query close to the visual baseline when audio is likely to be distracting.

Together, C1 decides *how much* audio to keep, while C2 decides *which* audio to attend to. This separation is important: the gating function changes the strength of fusion, whereas the injected query changes the content used to retrieve audio evidence. In the dog example, C1 can reduce the overall influence of sound for a visually focused query, while C2 can steer attention toward barking-related audio when the text explicitly asks for it.

Training objective. The final fused video representation is compared with the text representation using the same retrieval objective as the AVIGATE baseline. For a mini-batch of paired videos and captions, video-to-text and text-to-video similarities are optimized with a bidirectional contrastive loss, while the CLIP encoders and audio fusion modules follow the baseline training recipe. This keeps the experimental difference focused on the two query-aware fusion components.

3 Experiment

Setup. Evaluation is conducted on MSR-VTT [12] (9K train / 1K test JSFUSION split) and VATEX [11], reporting Recall@ K , Median Rank (MdR), and Mean Rank (MnR). Unless noted, all experiments use CLIP ViT-B/32 with text, visual, and audio modalities. Recall@ K measures whether the correct item appears in the top K retrieved results, while MdR and MnR measure the ranking position of the ground-truth match. Models train with distributed data parallelism for 5 epochs at effective batch size 512, learning rate $1e-4$, CLIP coefficient $1e-3$, warmup 0.1, learning-rate decay 0.9, 12 frames / 32 words, and seed 42. Cross-attention and audio query layers follow the AVIGATE baseline so that gains mainly reflect query-aware fusion rather than changes in the training protocol.

Comparison with the state of the art. On MSR-VTT (Table 1), Text-Guided AVIGATE reaches **66.0** R@1, improving over GAIS [14] by **+9.0** and over the AVIGATE baseline [4] by **+15.8**. The gains also hold at R@5 (+5.8 over GAIS) and R@10 (+3.3), while MnR drops from 7.6 to 3.3, indicating that query-aware fusion improves the ranking distribution rather than only a few top-1 cases. The contrast with AVIGATE is especially informative: adding audio with text-agnostic gates is not enough; the model must decide audio relevance from the query. The table also shows that prior audio-using methods do not automatically dominate vision-only methods. For example, AVIGATE and T-MASS both report 50.2 R@1 despite using different modalities, which supports the claim that the key issue is not simply whether audio is present, but whether the audio is selected according to the text query.

Table 1: MSR-VTT [12] text-to-video retrieval. Best in **bold**; strongest prior baseline underlined.

Method	Modality	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
CLIP4Clip [12]	V+T	43.1	70.4	80.8	2.0	15.3
ECLIPSE [5]	A+V+T	44.2	71.3	81.6	2.0	15.0
BridgeFormer [11]	V+T	44.9	71.9	80.3	2.0	15.3
X-CLIP [8]	V+T	46.1	73.0	83.1	2.0	13.2
X-Pool [2]	V+T	46.9	72.8	82.2	2.0	14.3
TS2-Net [6]	V+T	47.0	74.5	83.8	2.0	13.0
TEFAL [3]	A+V+T	49.4	75.9	83.9	2.0	12.0
CLIP-ViP [13]	V+T	50.1	74.8	84.6	1.0	—
AVIGATE [4]	A+V+T	50.2	74.3	83.2	1.0	13.8
T-MASS [10]	V+T	50.2	75.3	85.1	1.0	11.9
GAIS [14]	A+V+T	<u>57.0</u>	<u>83.1</u>	<u>90.9</u>	1.0	<u>7.6</u>
Proposed (ViT-B/32)	A+V+T	66.0	88.9	94.2	1.0	<u>3.3</u>
Proposed (ViT-B/16)	A+V+T	67.5	90.6	95.1	1.0	3.1

Ablation study. Table 2 removes each contribution in turn on MSR-VTT (ViT-B/32). Removing C1 or C2 lowers T2V R@1 from 66.0 to 64.4 and 64.3, respectively (Table 2). The drops are similar in size, which indicates that the two components are not redundant: C1 changes the strength of audio fusion, whereas C2 changes the query used to retrieve audio evidence. V2T is more sensitive: full-model R@1 is **37.5**, versus 21.5 without C1 and 25.6 without C2. Since V2T matches one fused video representation against many candidate captions, it is highly sensitive to whether the representation preserves query-level semantics; the large V2T gap therefore supports the claim that the text signal is doing more than simply adding another modality.

Table 2: Ablation on MSR-VTT (CLIP ViT-B/32).

Configuration	Text-to-Video			Video-to-Text		
	R@1	R@5	R@10	R@1	R@5	R@10
w/o Text-Conditioned Gating (C1)	64.4	89.8	94.2	21.5	53.9	72.9
w/o Text-Injected MHA Query (C2)	64.3	89.3	94.1	25.6	61.8	78.8
Full Model (C1 + C2)	66.0	88.9	94.2	37.5	72.8	86.2

Effect of the visual backbone. The last two rows of Table 2 compare two CLIP visual backbones. Swapping ViT-B/32 for the higher-resolution ViT-B/16 raises MSR-VTT R@1 from 66.0 to **67.5**, improves R@5/R@10 to 90.6/95.1, and reduces MnR from 3.3 to 3.1. Thus the query-aware fusion does not saturate with a stronger visual encoder. Instead, finer visual tokens appear to provide a better anchor for the Text-Injected MHA Query when it aligns audio evidence to the text.

Cross-dataset generalization. On VATEX (Table 3), zero-shot transfer is already strong: the MSR-VTT-trained model reaches 73.7 R@1 without VATEX training. After fine-tuning, R@1 increases to 80.5 with ViT-B/32 and 85.5 with ViT-B/16. The same ViT-B/16 > ViT-B/32 ordering appears on both datasets, suggesting that the method learns a transferable query-aware fusion rule rather than overfitting to MSR-VTT captions. The high R@5 and R@10 values on VATEX also indicate that remaining errors are often near misses rather than complete retrieval failures.

Table 3: VATEX [11] text-to-video results on the 10% test subset.

Setting	Backbone	R@1 ↑	R@5 ↑	R@10 ↑	MdR ↓	MnR ↓
Zero-shot	ViT-B/32	73.7	92.5	95.3	1.0	2.5
Fine-tuned	ViT-B/32	80.5	97.3	98.9	1.0	1.6
	ViT-B/16	85.5	98.0	99.2	1.0	1.6

Discussion. Across experiments, the largest gains appear where fine-grained semantic precision matters most: R@1, V2T retrieval, and transfer to a new dataset. This supports the central claim that audio relevance is query-dependent and should be controlled by text rather than by video-only fusion gates. The ablation also clarifies the roles of the two proposed components. Text-conditioned gates are responsible for regulating the overall amount of audio information, while the text-injected query determines the semantic direction of the audio lookup. This division is useful because not every query needs audio, and among queries that do need audio, not every sound in the clip is relevant.

4 Conclusion

This report presented Text-Guided AVIGATE, which makes audio-visual fusion for text-to-video retrieval *query-aware* through a Text-Conditioned Gating Function and a Text-Injected MHA Query. The method improves R@1 by +15.8 over the AVIGATE baseline and +9.0 over GAIS on MSR-VTT, remains effective with a stronger CLIP backbone, and transfers well to VATEX. Ablations confirm that text-conditioned gating and gated text-query injection provide complementary benefits: one controls the amount of audio fusion, and the other makes the audio attention query semantically aware. Overall, the results support the view that audio-visual retrieval should treat audio relevance as a query-dependent decision rather than as a fixed property of the video.

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